Modelling TB in Viet Nam with the EndTB model

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EndTB model

The **EndTB model** is a dynamic simulation model of Mtb transmission and TB progression in a population structured by age, TB stage, HIV status, RISK, SES, TB Vaccine status (developed by Roel Bakker)

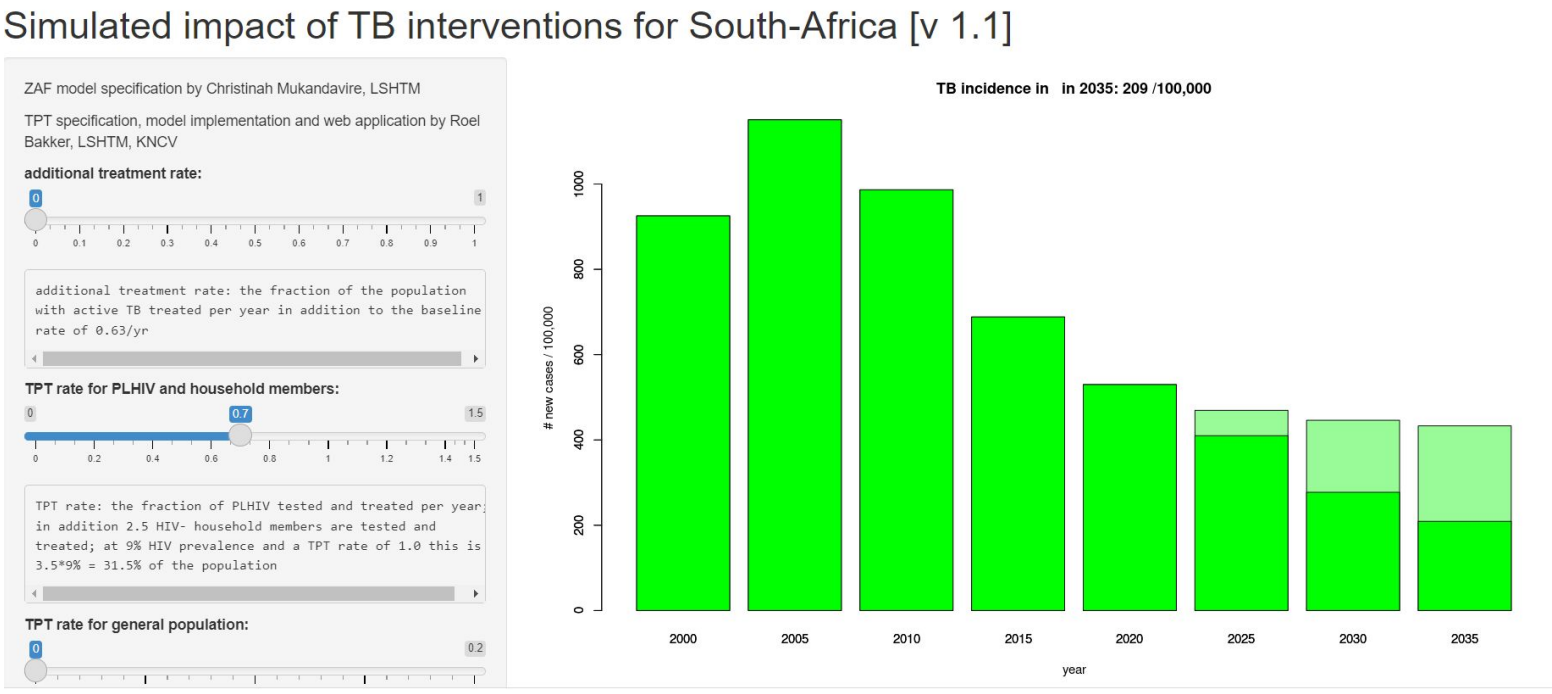
EndTB is based on the **TBvax** model that is used by the TB Modelling group (prof Richard White) at LSHTM since 2020 in a project funded by WHO and BMGF to estimate the impact of TB **vaccines** on Mtb transmission and TB progression

The **model specification** is in the appendix of a paper submitted to Lancet Global Health (and available on MedrXiv now at The impact of alternative delivery strategies for novel tuberculosis vaccines in low- and middle-income countries: a modelling study | medRxiv)

User friendly web version of EndTB

https://roelb54.shinyapps.io/TBvaxdemo/

[may be slow in starting up and takes 10-15 secs for a simulation run]

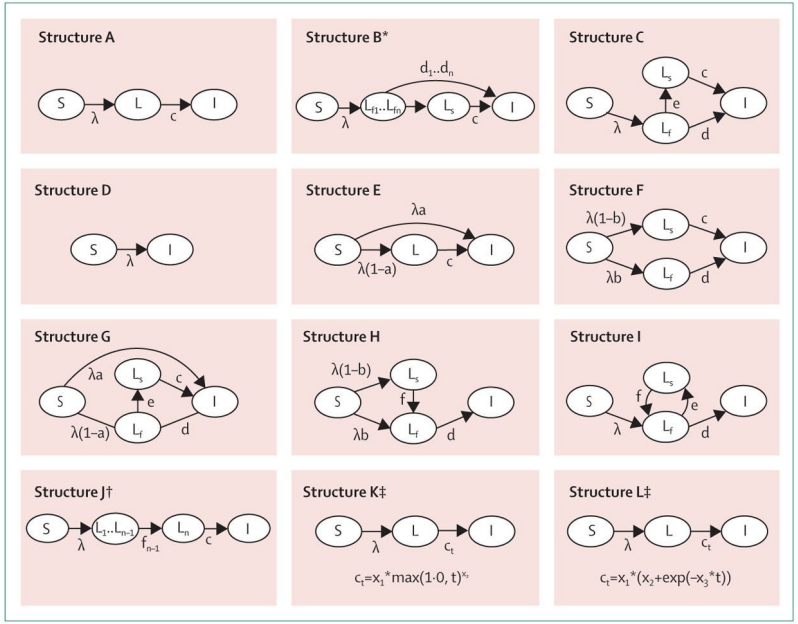


Why EndTB?

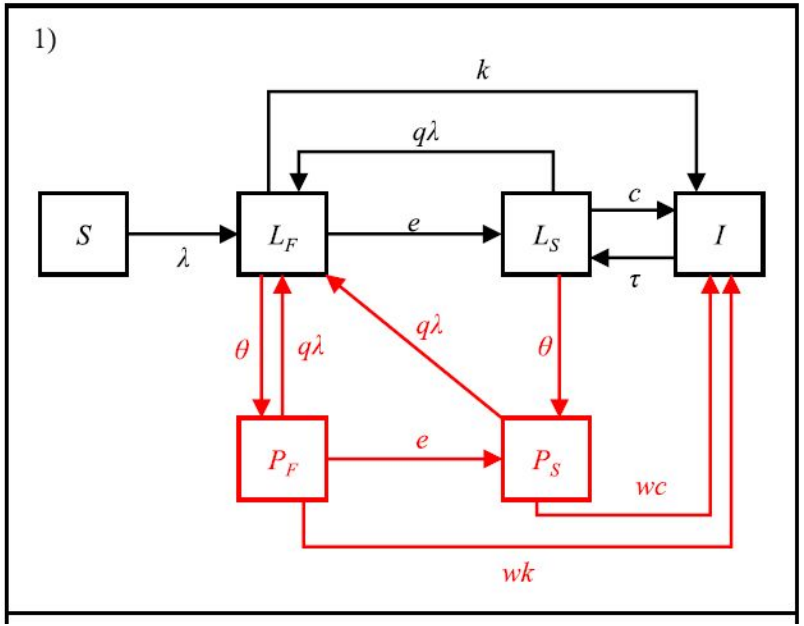
- make using transmission models more accessible to in-country modellers / epidemiologist
- (for now) expert modellers will do calibration (parameter estimation using data)
- train in-country modellers
- pilot this approach in ?
- develop e-learning
- extend to other countries
- collaboration with TB Modelling Group at LSHTM

EndTB model design - 1. background

Many assumptions about TB natural history



[from the 2018 paper by Nick Menzies]



[from the 2020 paper by Sumner and White on TPT]

EndTB model design - 2

Model requirements

- flexible definition (not hard coded) of TB natural history
- as most HIV deaths are due to TB:
  - include HIV (undiagnosed, diagnosed, on ART, ART suppressed) natural history

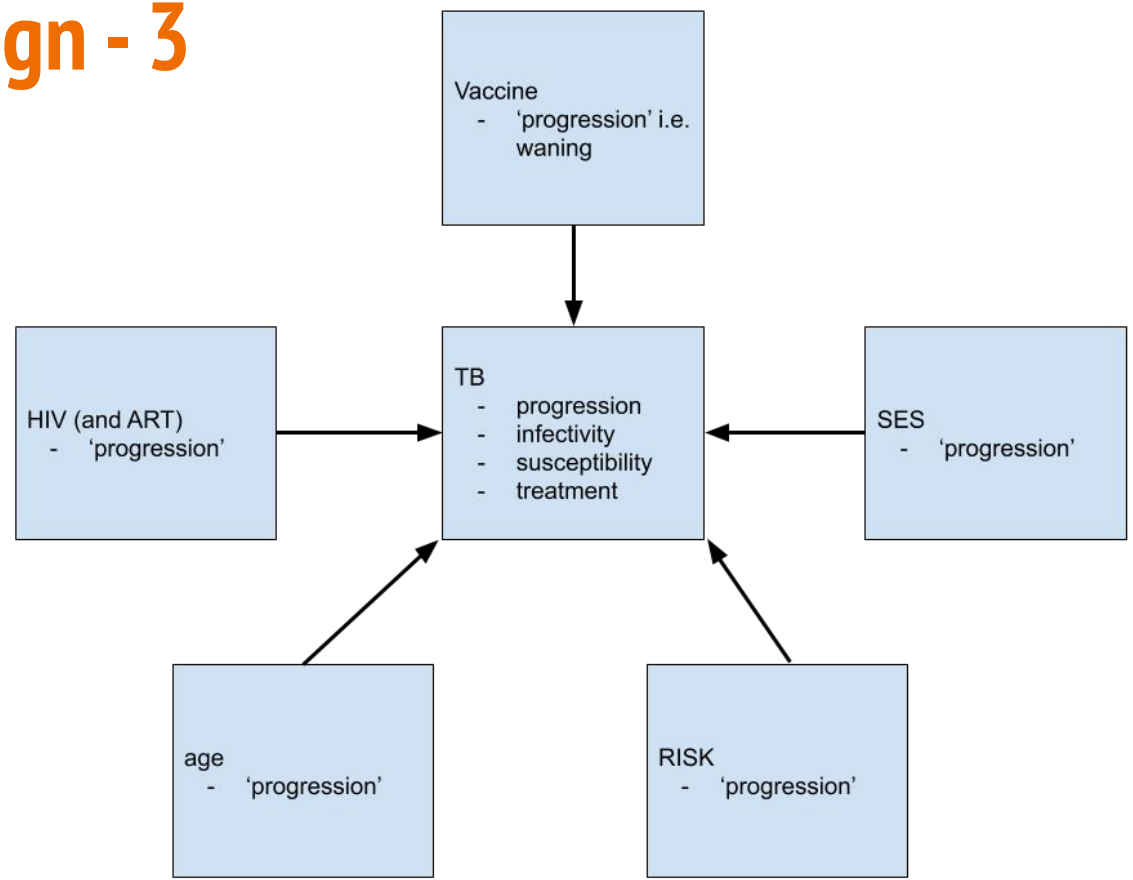
How to include HIV?

- as a separate ‘dimension’ i.e. to enable simple specification of HIV natural history
- enhanced TB progression in HIV+ by using HIV dependent parameters

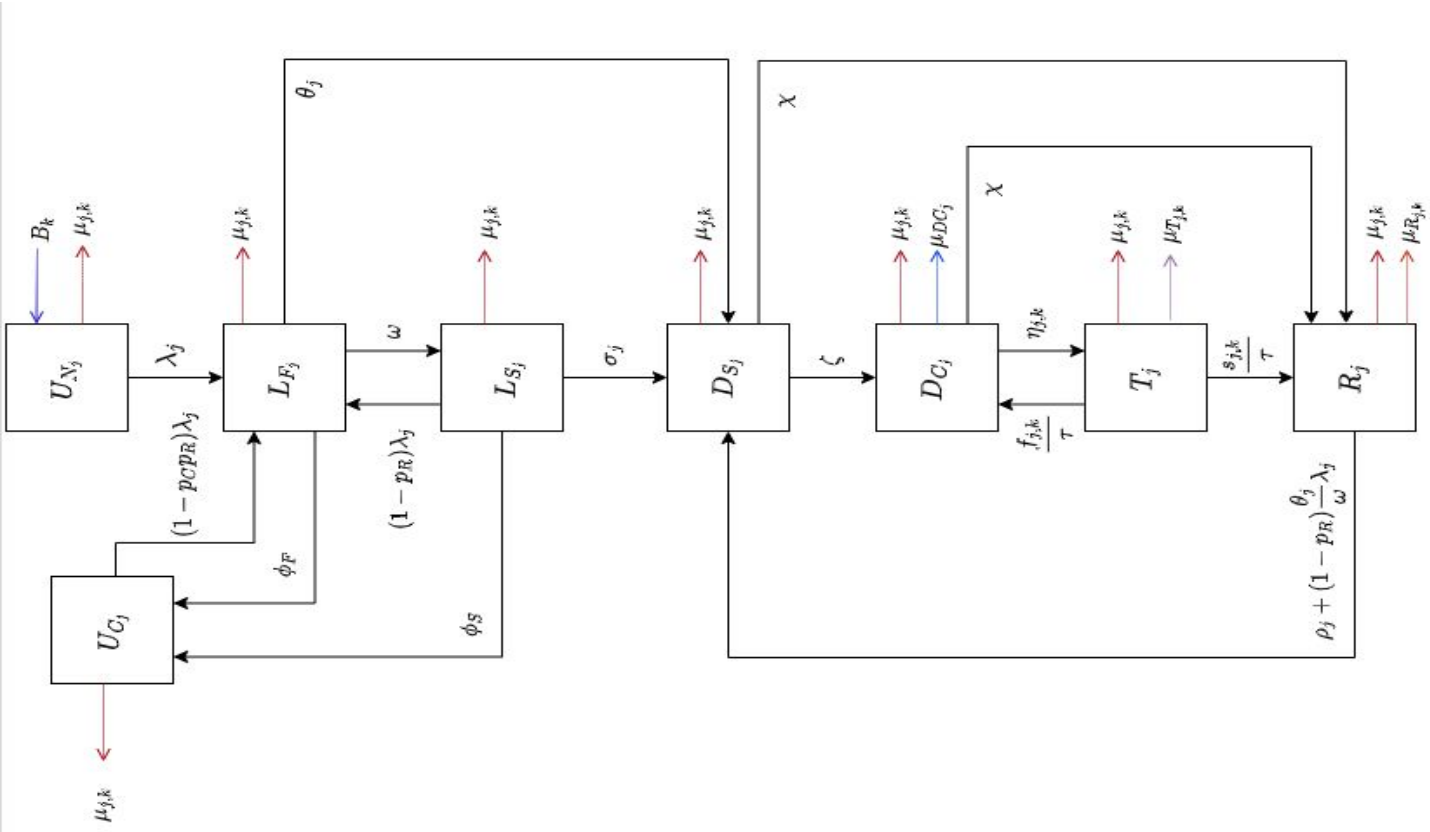
EndTB model design - 3

The full TB model includes options to define **dependencies** (see right panel), so any TB parameter may depend on any of:
age, **HIV**, **RISK**, **SES**, and **Vx**.

(and even reverse dependencies are possible)



EndTB: example TB natural history (from R. Clark LMIC countries)



example TB natural history definition

```
<TB>
  <TB.stages>
    <stage name="Un" fraction.at.birth="1"/>
    <stage name="Uc"/>
    <stage name="Lf"/>
    <stage name="Ls"/>
    <stage name="Ds"/>
    <stage name="Dc"/>
    <stage name="T"/>
    <stage name="R"/>
    <stage name="TBdead"/>
    <stage name="TBHIVdead"/>
    <stage name="Rdead"/>
    <stage name="RHIVdead"/>
  </TB.stages>
```

example TB state transition definition

```
<transition.matrix>
  <transition from="Lf" to="Ls" rate="omega"/>
  <transition from="Lf" to="Uc" rate="phiF"/>
  <transition from="Ls" to="Uc" rate="phiS"/>
  <transition from="Ds" to="Dc" rate="zeta*(1+atheta*th1)"/>
  <transition from="Ds" to="R" rate="chi/(1+aHIVslowrecov*th1)"/>
  <transition from="Dc" to="R" rate="chi/(1+aHIVslowrecov*th1)"/>
  <transition from="Dc" to="TBdead" rate="sage*muDc*(!HIVpos)"/>
  <transition from="Dc" to="TBHIVdead" rate="sage*muDc*(1+aHIVdeath*HIVx)*HIVpos"/>
  <transition from="R" to="Rdead" rate="sage*muK*(!HIVpos)*0.22"/>
  <transition from="R" to="RHIVdead" rate="sage*muK*(1+aHIVdeath*HIVx)*HIVpos*0.22"/>
</transition.matrix>
```

EndTB model in summary

User benefits

- the model is a multi-purpose **tool** for infectious disease dynamics
- the user defines the model
  - age groups
  - TB natural history and transmission
  - dependencies by HIV, RISK, SES and TB Vaccine
  - age mixing (contact matrices based on PolyMod study)

Technical benefits

- simple and fast due to matrix operations

Modelling Viet Nam - 1

TB Natural History

- relatively simple: Un, Uc, Lf, Ls, Ds, Dc, T, R, TBdead, Rdead

Mixing

- assortative mixing by sex (in addition to mixing by age [PolyMod study])

Data (WHO)

- notifications by sex (1996 thru 2015) and by sex and age group (2016 thru 2023)
- estimated incidence (2000 thru 2023)
- TB deaths (2000 thru 2023)
- prevalence by age group and 15-99 by sex (2007)
- prevalence by sex and age group (2018)
- 229 data points (i.e. targets) to be fitted

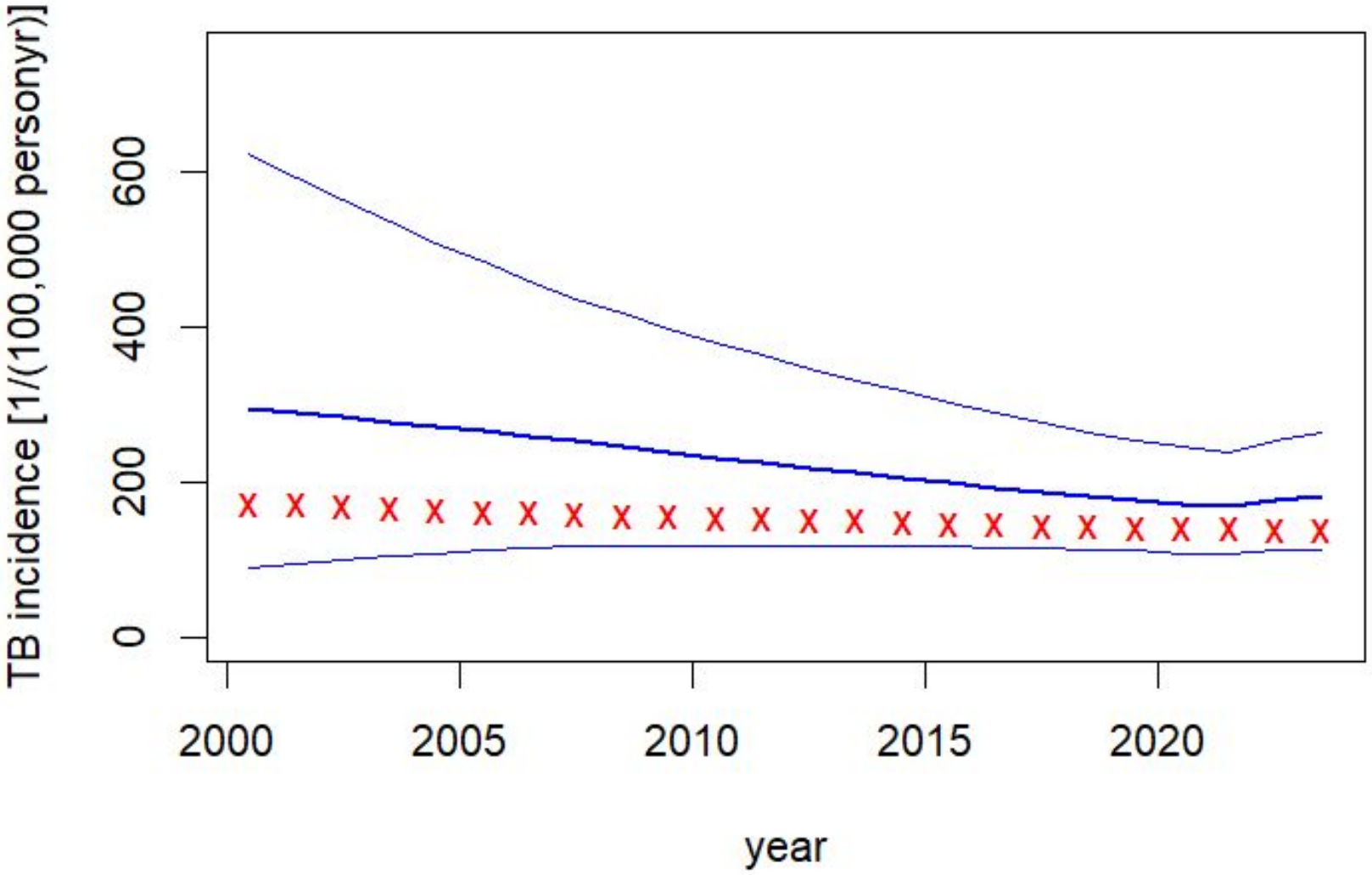
Modelling Viet Nam - 2

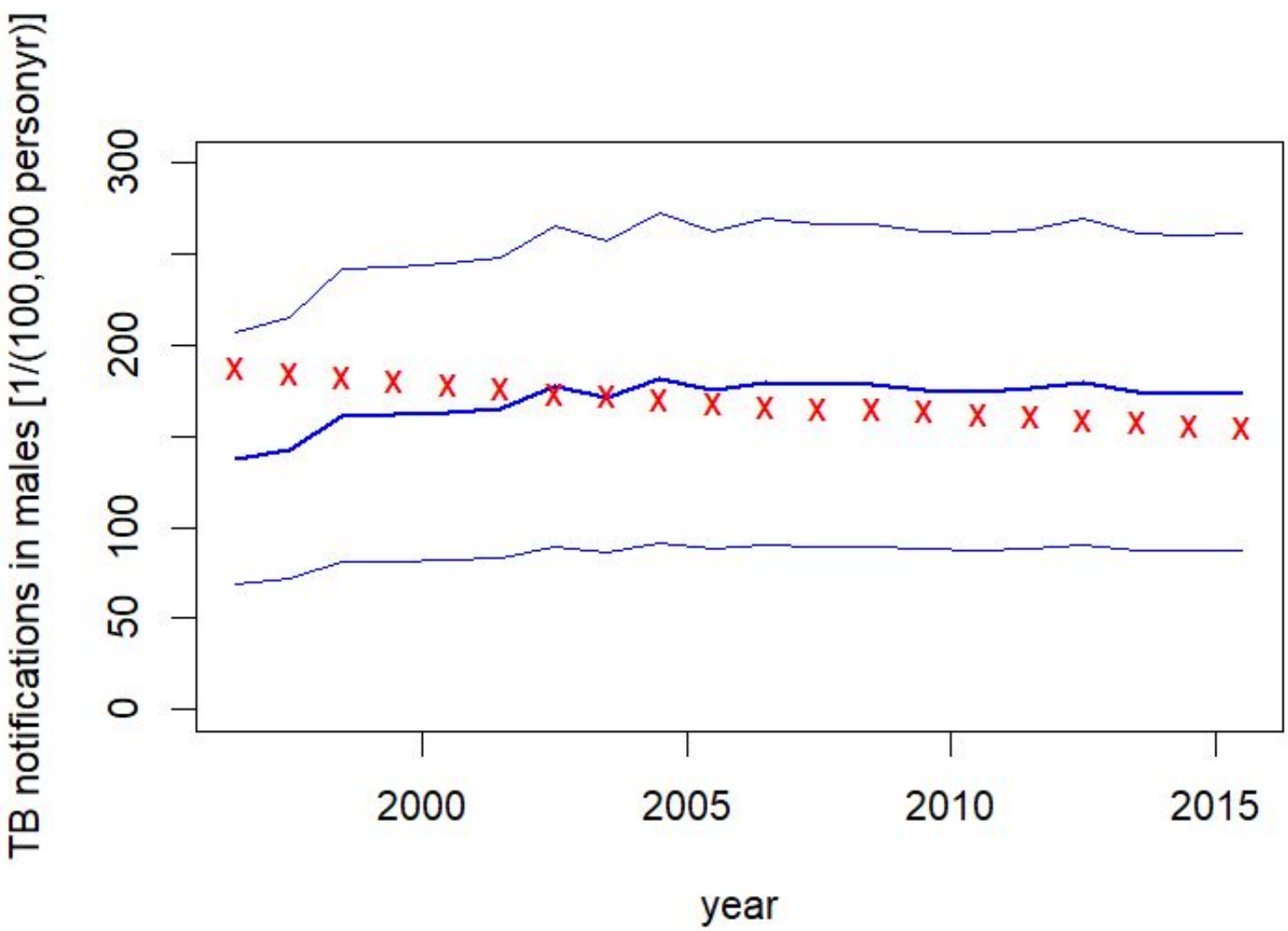
25 model parameters were fitted

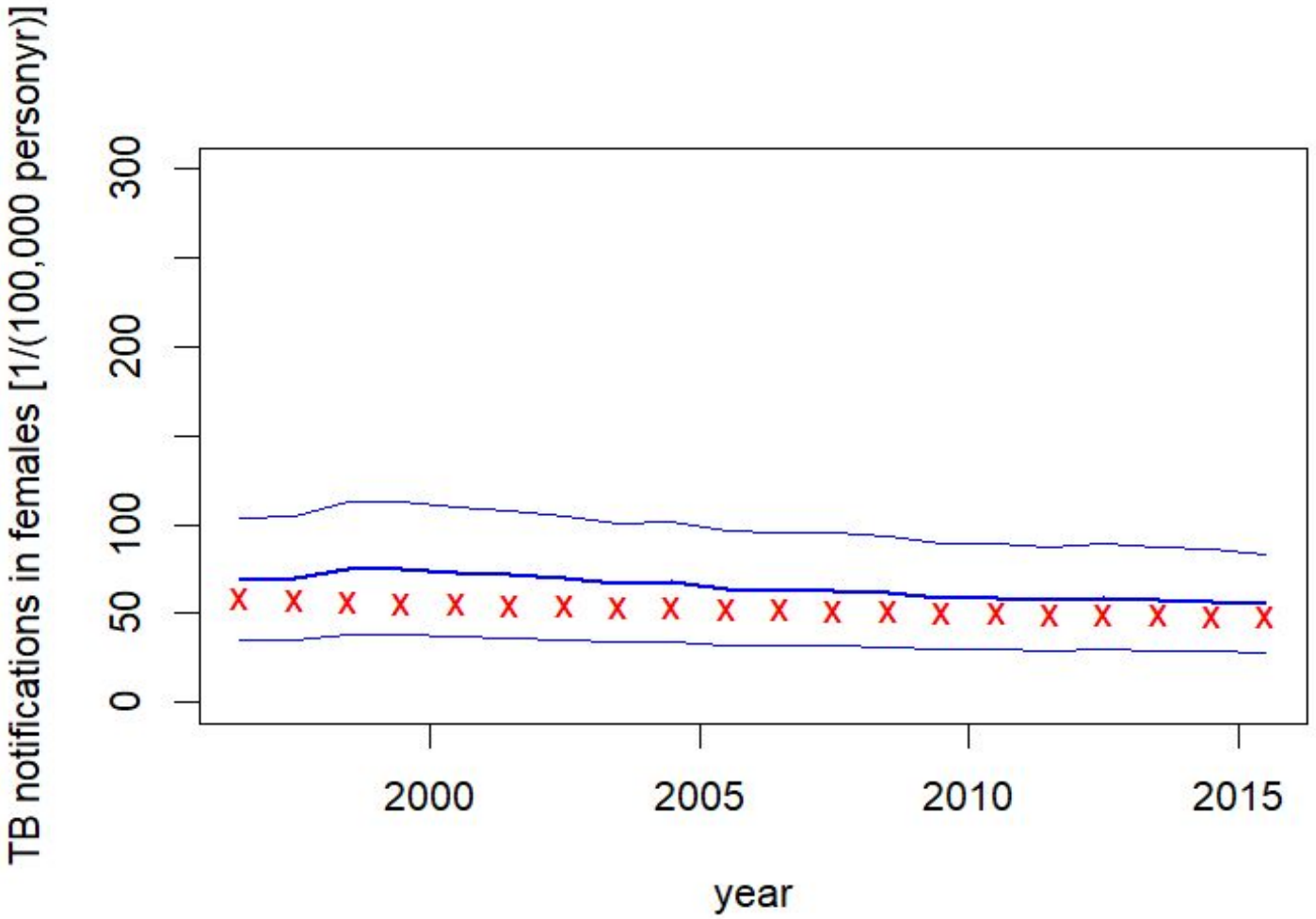
- using optimization (Levenberg-Marquardt method) to minimize the weighted sum of squares of the differences between targets (data) and model values
- note that set of the parameter values may be unique
- future: use a Bayesian method e.g. ABC (approximate Bayesian computing) or hmer (history matching)

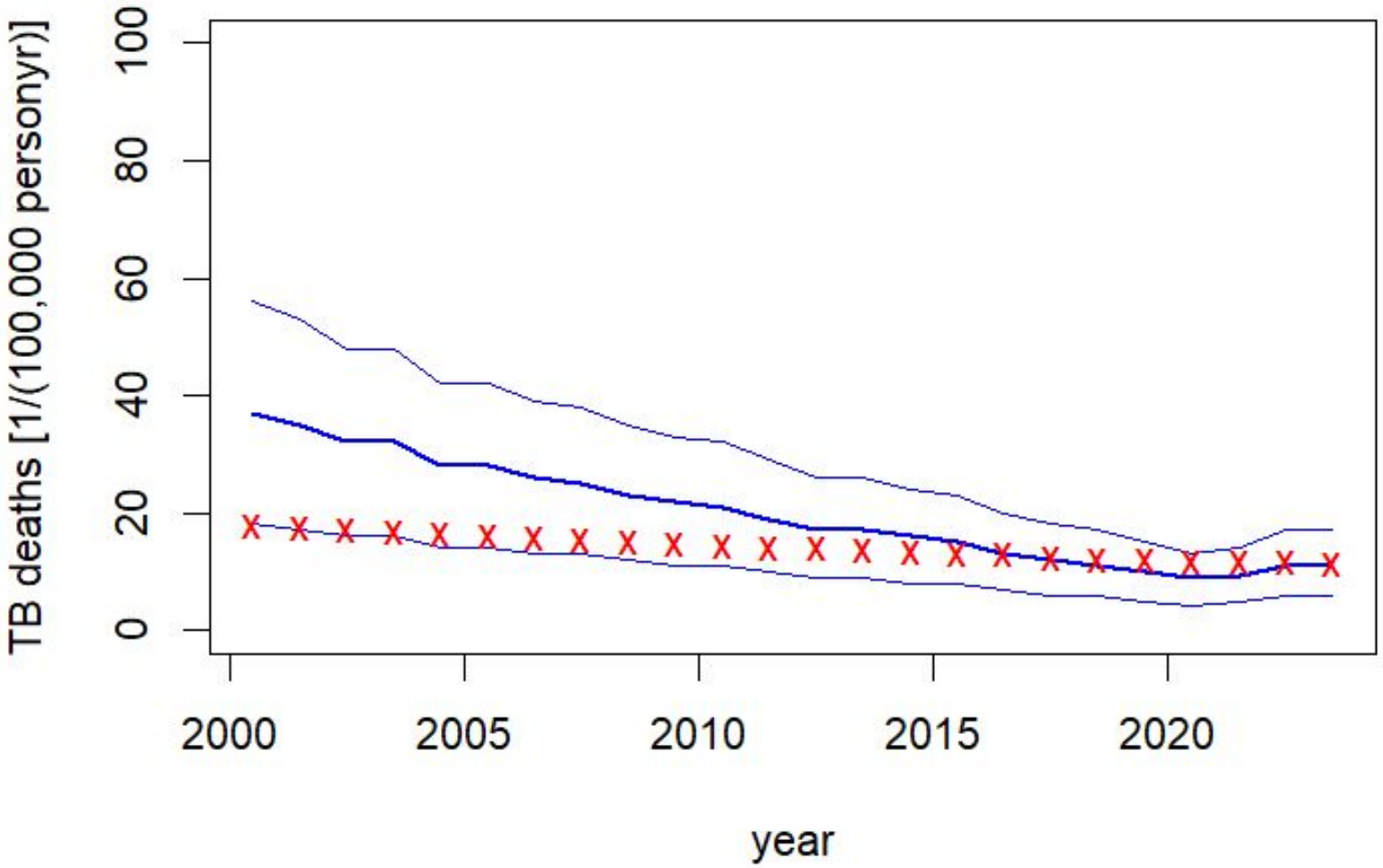
sex dependent parameters

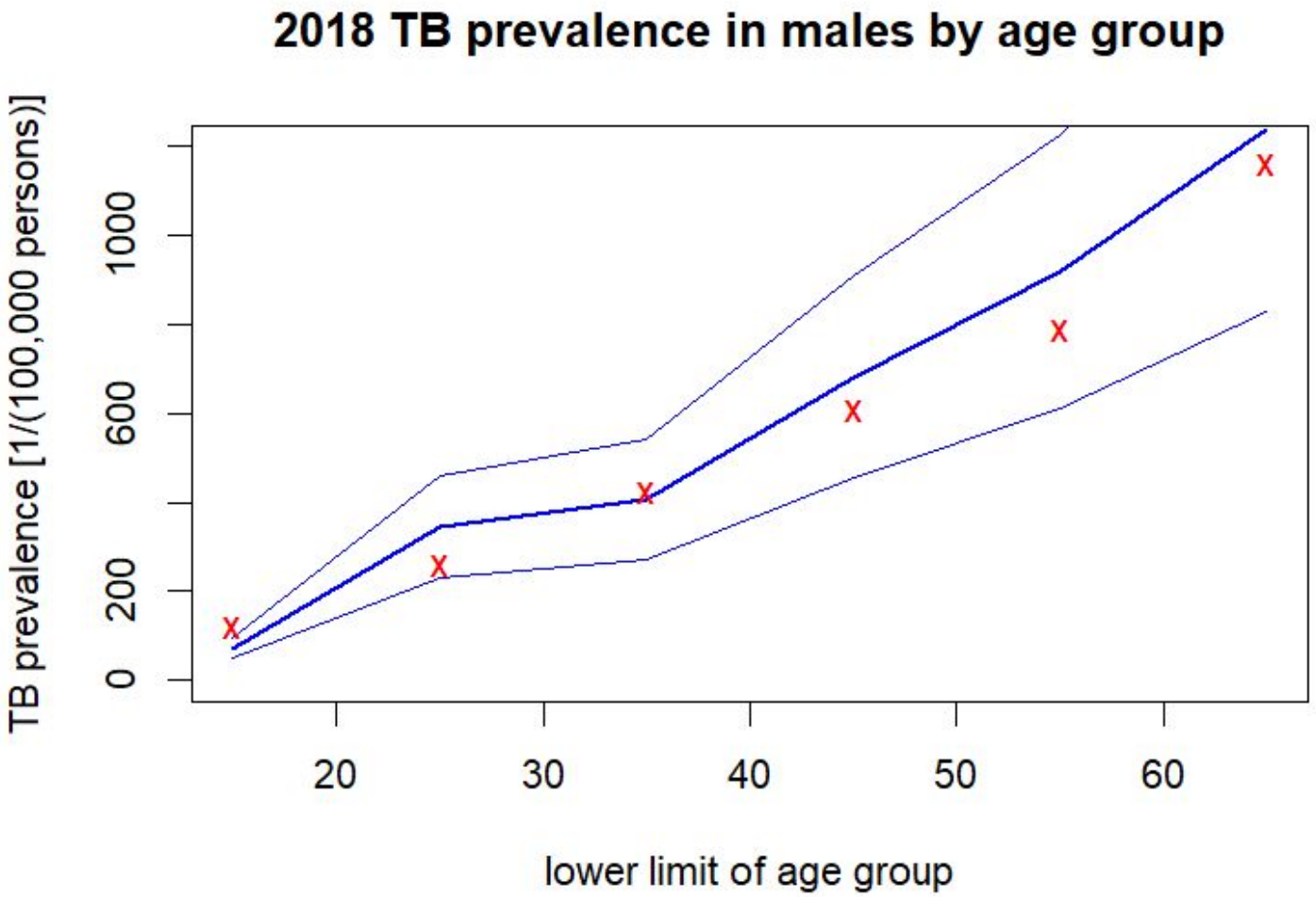
- assortativity (0.4) in contacts between M-M and F-F vs M-F and F-M (with 0 representing proportional mixing and 1 100% assortative mixing)
- rate of progression from Lf->Ds, Ls->Ds, R->Ds (F: 60% of M rate)
- susceptibility to transmission (no difference)
- treatment success (no difference)

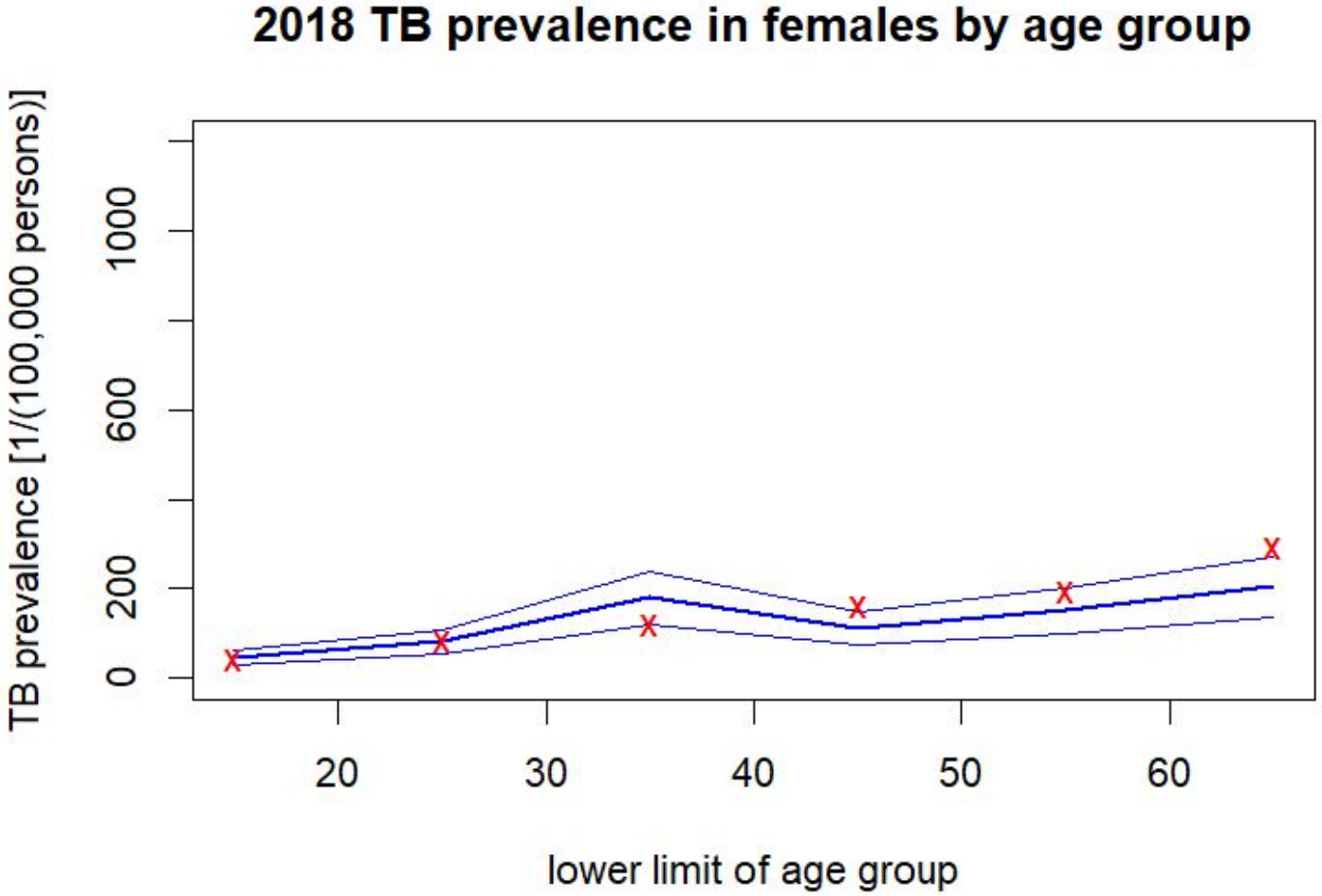

TB incidence [1/(100,000 personyr)]
0
200
400
600
2000
2005
2010
2015
2020
year

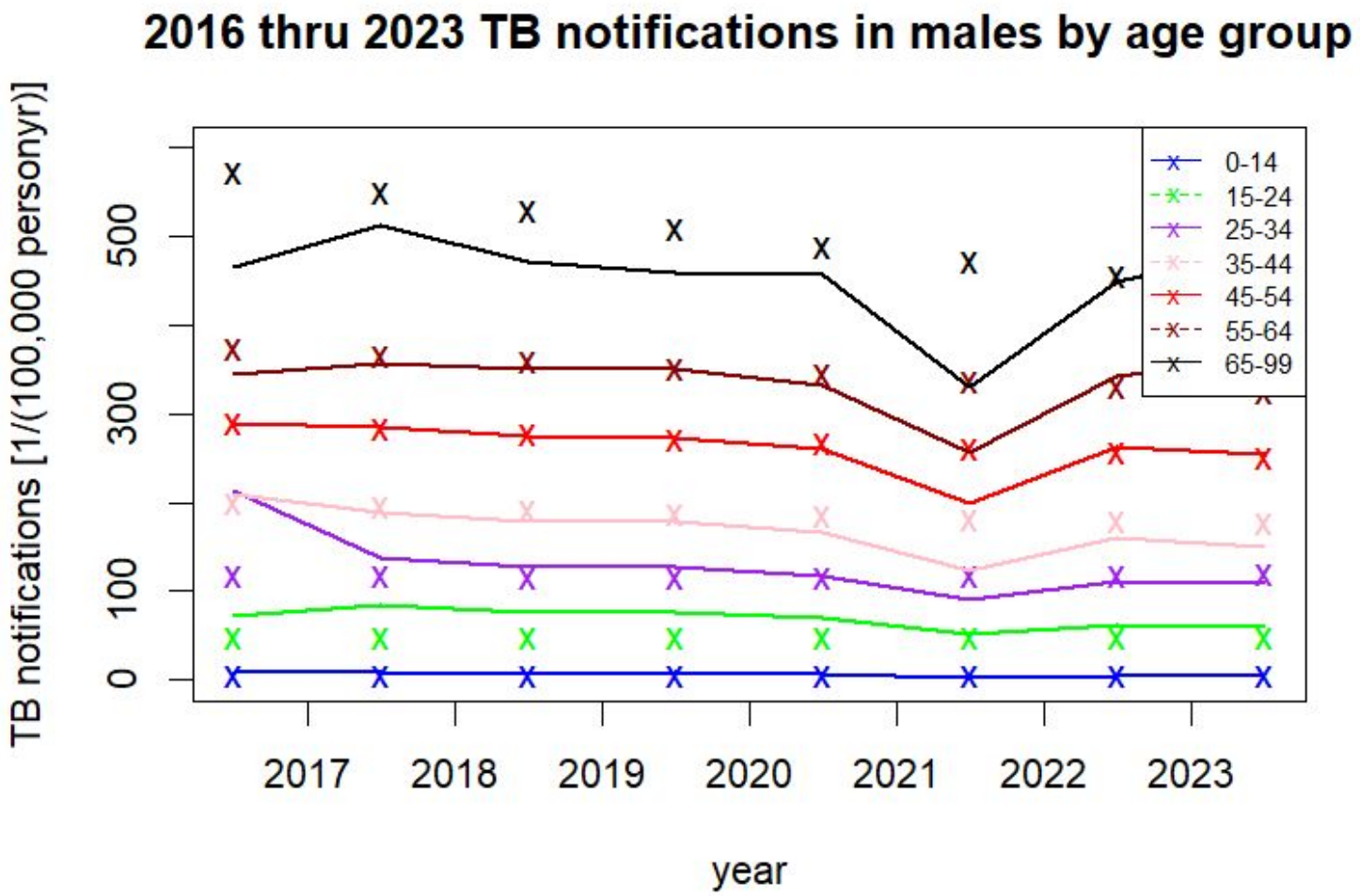

TB notifications in males [1/(100,000 personyr)]
0
50
100
200
300
2000
2005
2010
2015
year

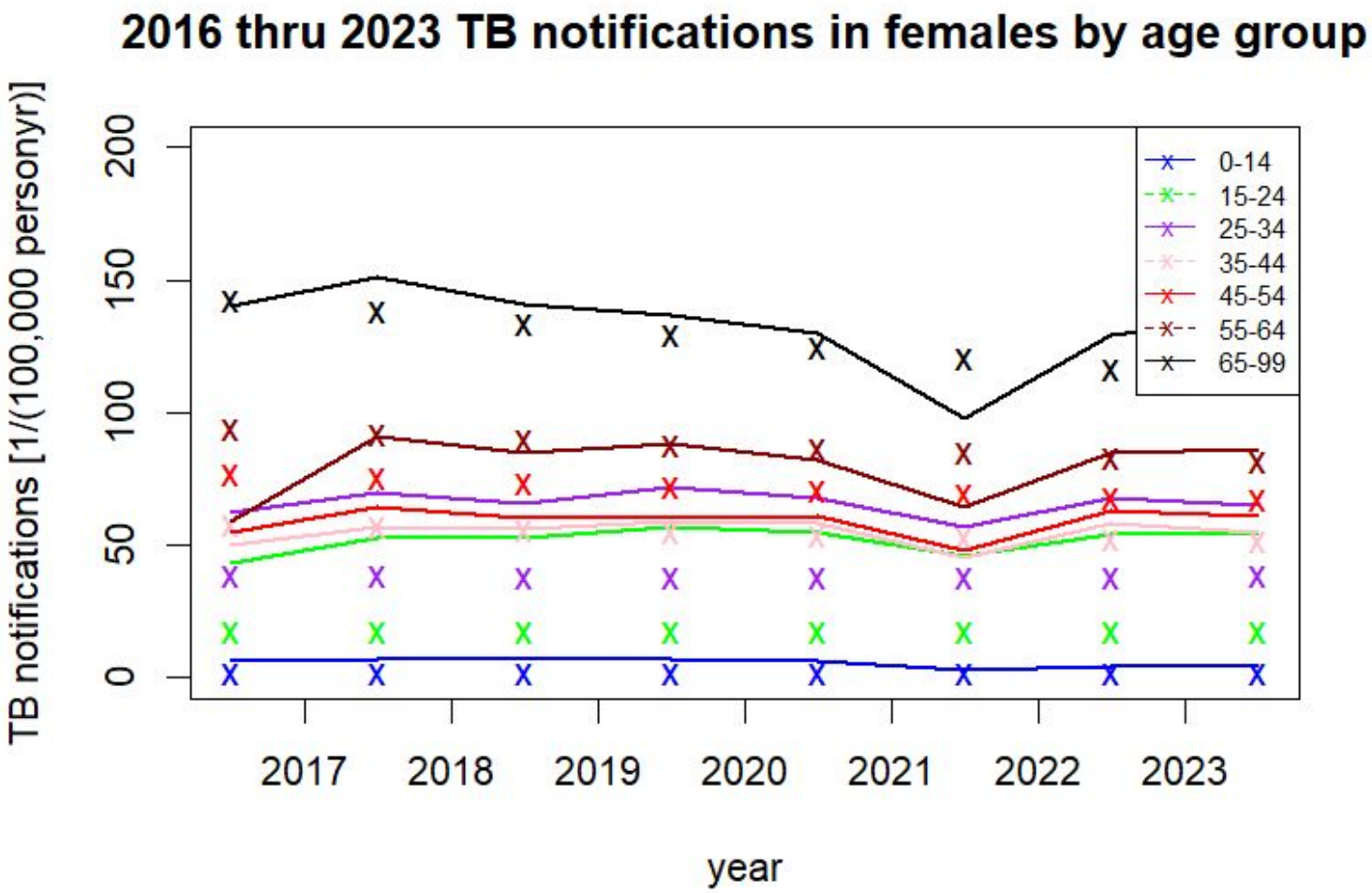

TB notifications in females [1/(100,000 personyr)]
0
50
100
200
300
2000
2005
2010
2015
year


TB deaths [1/(100,000 personyr)]
0
20
40
60
80
100
2000
2005
2010
2015
2020
year


2018 TB prevalence in males by age group
TB prevalence [1/(100,000 persons)]
0
200
600
1000
20
30
40
50
60
lower limit of age group


2018 TB prevalence in females by age group
TB prevalence [1/(100,000 persons)]
0
200
600
1000
20
30
40
50
60
lower limit of age group


2016 thru 2023 TB notifications in males by age group
TB notifications [1/(100,000 personyr)]
0
100
300
500
2017
2018
2019
2020
2021
2022
2023
year
0-14
15-24
25-34
35-44
45-54
55-64
65-99


2016 thru 2023 TB notifications in females by age group
TB notifications [1/(100,000 personyr)]
0
50
100
150
200
2017
2018
2019
2020
2021
2022
2023
year
0-14
15-24
25-34
35-44
45-54
55-64
65-99

Discussion / action points

- improve model fit?
  - TB incidence
  - TB deaths
  - TB notifications in F by age group
- confidence intervals on data?
- DR-TB not yet included
- HIV not included (though < 1% prevalence)
- additional data to be used? e.g. treatment success?
- natural history model used is not the most recent as published by LSHTM: https://www.pnas.org/doi/10.1073/pnas.2221186120
- which interventions to be modelled?
- develop web version

Questions?

Technical details - 1

Performance

- The rate of change (of all variables) can be calculated by simple **matrix multiplications**
- A simulation run of 150 years with a model with about 50,000 (82 age groups * 10 TB states * 2 SES * 10 HIV/ART * 3 Vx) state variables (i.e. 50,000 differential equations) requires just 20 - 40 seconds

Complexity

- Basically all that happens in the simulation are state transitions:
  - transmission (Un -> Lf)
  - progression (Lf -> Ls)
  - even incidence data can be expressed as state transitions
    - HIV negative -> HIV+
    - ART (HIV+ -> ART)
    - Vx implementation: (unvaccinated -> protected)

Technical details - 2

- In an SIR model defined by (where the dot on top of S, I and R indicate the time derivative i.e. dS/dt etc.):

$$\dot{S} = -\beta \cdot I \cdot S/N$$
$$\dot{I} = \beta \cdot I \cdot S/N - \gamma \cdot I$$
$$\dot{R} = \gamma \cdot I$$

with $\lambda = \beta \cdot I/N$

- The time derivatives of S, I and R can be calculated by:

$$\begin{bmatrix} \dot{S} \\ \dot{I} \\ \dot{R} \end{bmatrix} = \begin{bmatrix} -\lambda & 0 & 0 \\ \lambda & -\gamma & 0 \\ 0 & \gamma & 0 \end{bmatrix} \cdot \begin{bmatrix} S \\ I \\ R \end{bmatrix}$$

- i.e. a multiplication of a matrix and a vector (S, I, R)
  (that could just as well be a matrix with age groups in columns)
- this is the basis for all calculations in the TB Vx model (though the matrices are larger e.g. 648 rows x 648 columns for parameter matrices and 648 rows by 82 columns for the state variables